

## Local Binary Patterns (LBP) & Histogram of Oriented Gradient (HoG)

# Local Binary Patterns (LBP)

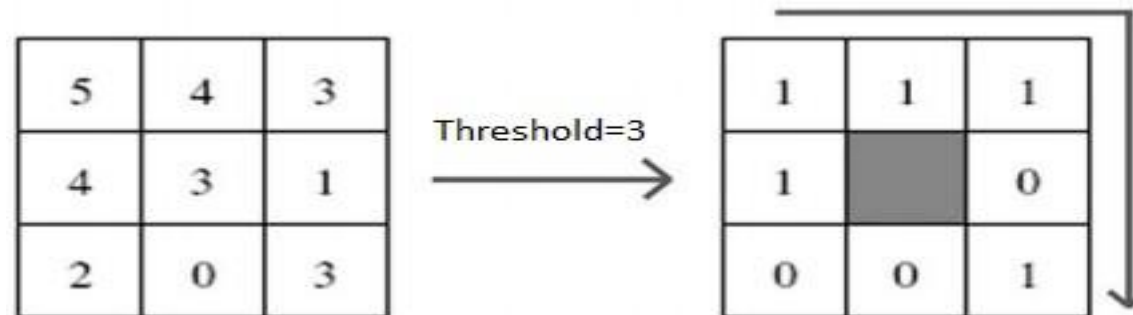
# LBP

- Mainly designed for monochrome still images
  - Have been extended for color (multi channel)
  - Videos ...
- Were introduced by Ojala et al. *“A comparative study of texture measures with classification based on feature distributions”*. *Pattern Recognition*. **29(1)**, 51–59 (**1996**)

# LBP

- The local binary pattern operator is an image operator which **transforms** an image into an array or image of integer labels describing small-scale appearance (textures) of the image.
- These labels directly or their statistics are used for further analysis.

- It is assumed that a texture has locally two complementary aspects, a pattern and its strength
- local binary pattern operator works in a 3×3 pixel



- The pixels in this block are
  - thresholded by its center pixel value,
  - multiplied by powers of two (Decimal)
  - then summed to obtain a label for the center pixel
  - 256 different labels



# 1. Local Binary Pattern (LBP)

- Description of pixels neighbourhood
- Binary short code to describe neighbourhood
- Operates by taking difference of central pixel with neighbouring pixels
- Mathematically

$$LBP_{R,P} = \sum_{p=0}^{P-1} \left( g_p - g_c \right) \cdot 2^p$$

where,

neighborhood pixels ( $g_p$  in each block

is the response of the center pixel value  $g_c$  for a 3x3 cell, where  $P = 8$ )

$R$  radius (for 3x3 cell, it is 1).

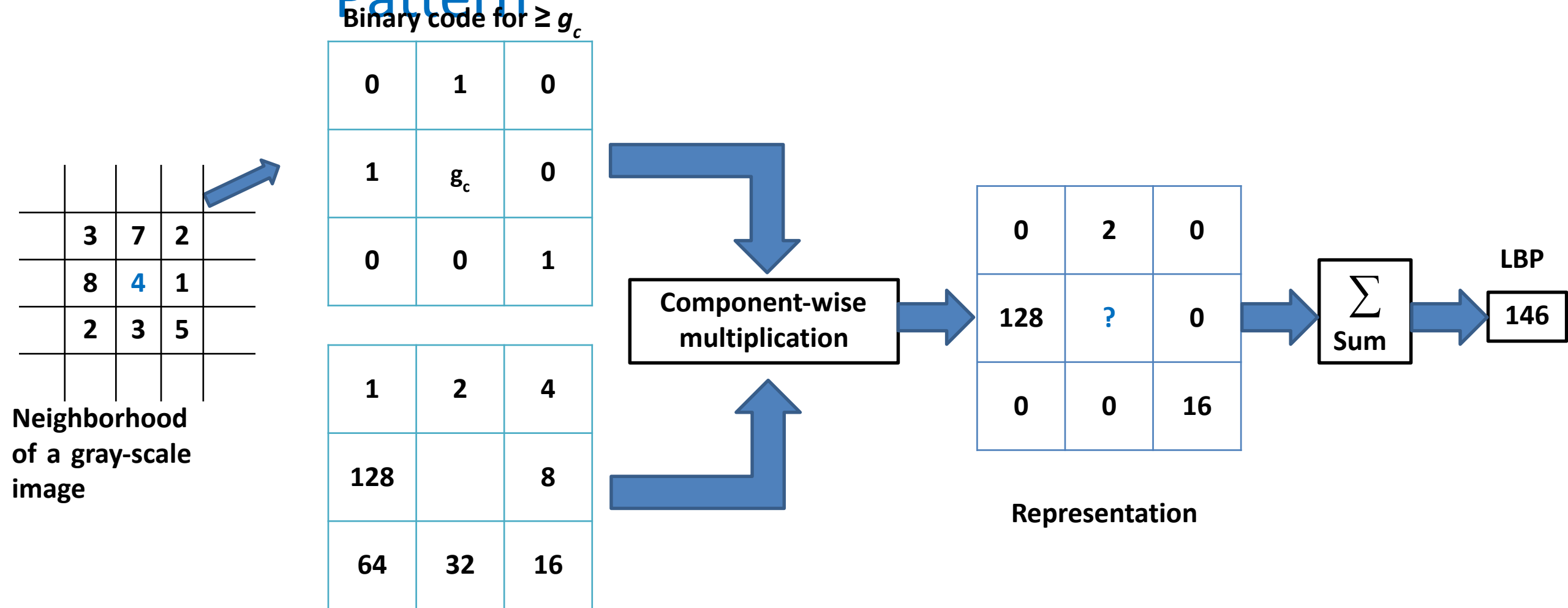
Coordinates of " $g_c$ " is (0,0) and of " $g_p$ " is  $(x + R\cos(2\pi p/P), y - R\sin(2\pi p/P))$

Binary threshold function  $s(x)$

$$s(x) = \begin{cases} 0, & x < 0 \\ 1, & x \geq 0 \end{cases}$$

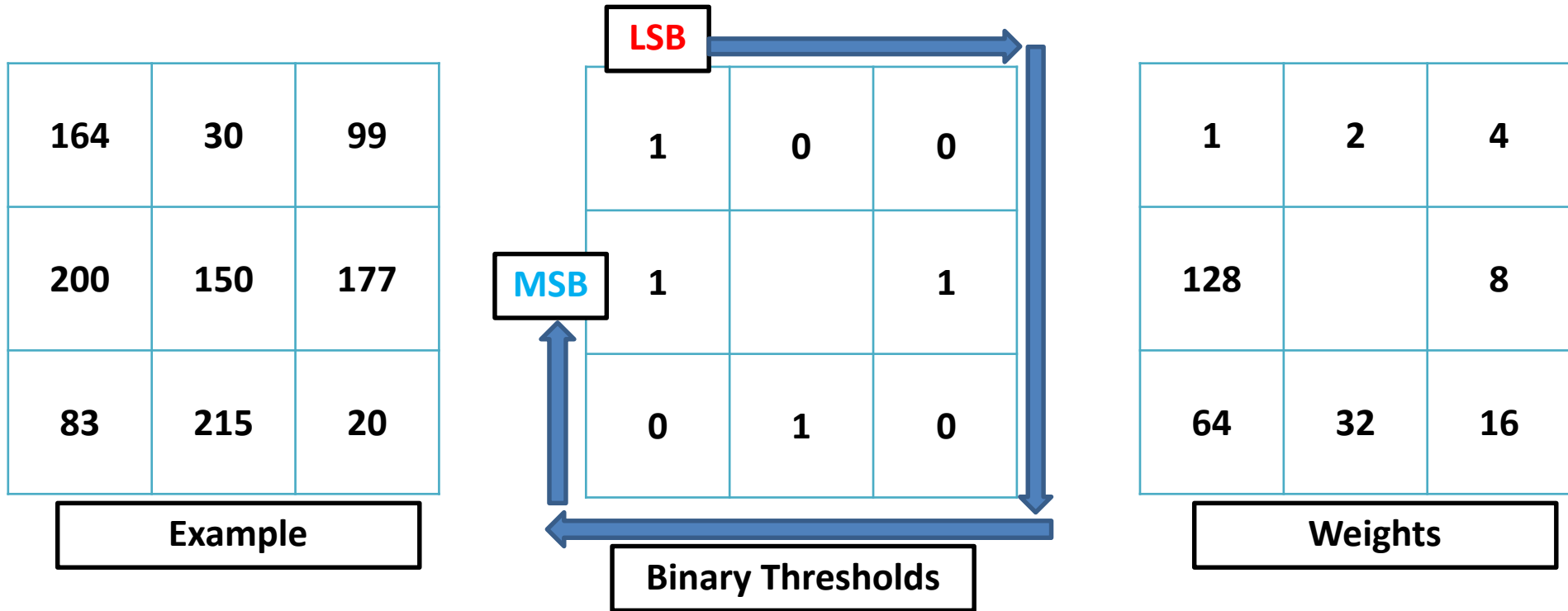


# Computation of Local Binary Pattern



Example of how the *LBP operator* works

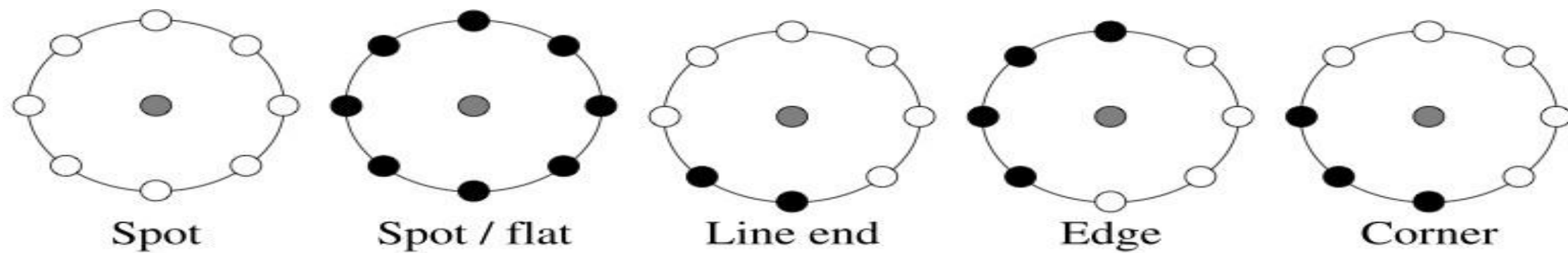
# Computation of LBP



|                 |            |   |   |   |   |   |   |            |
|-----------------|------------|---|---|---|---|---|---|------------|
| Binary Pattern: | 1<br>(MSB) | 0 | 1 | 0 | 1 | 0 | 0 | 1<br>(LSB) |
|-----------------|------------|---|---|---|---|---|---|------------|

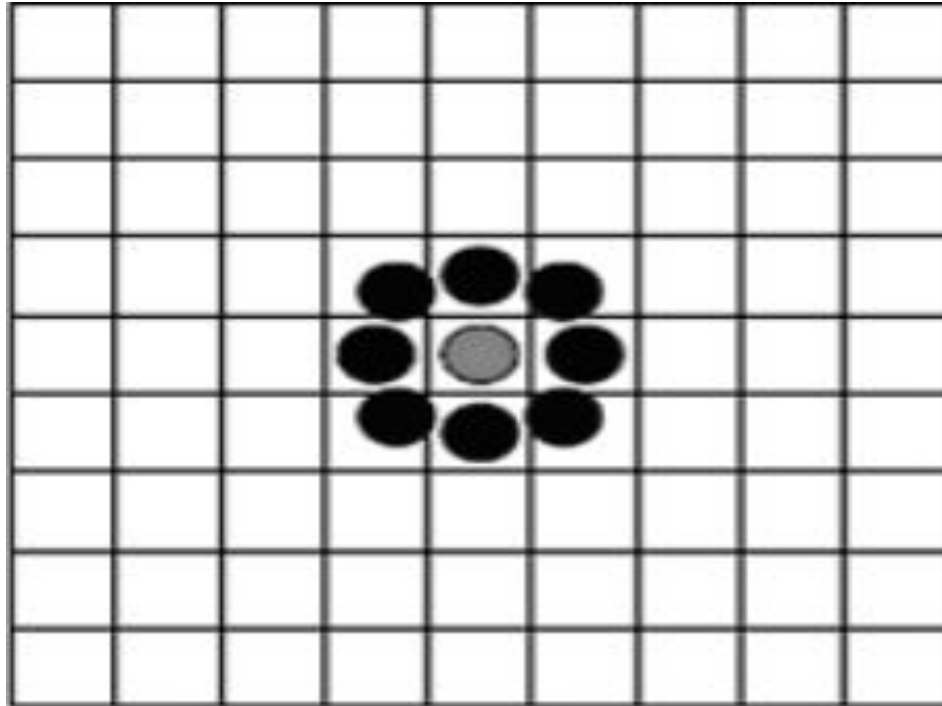
|                        |                |                |                |                |                |                |                |                |
|------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| Code/Weight ( $2^p$ ): | $1 \times 2^7$ | $0 \times 2^6$ | $1 \times 2^5$ | $0 \times 2^4$ | $1 \times 2^3$ | $0 \times 2^2$ | $0 \times 2^1$ | $1 \times 2^0$ |
|                        | = 128          | = 0            | = 32           | = 0            | = 8            | = 0            | = 0            | = 1            |

|      |                                          |
|------|------------------------------------------|
| LBP: | $1 + 0 + 0 + 8 + 0 + 32 + 0 + 128 = 169$ |
|------|------------------------------------------|



**Fig. 2.3** Different texture primitives detected by the LBP

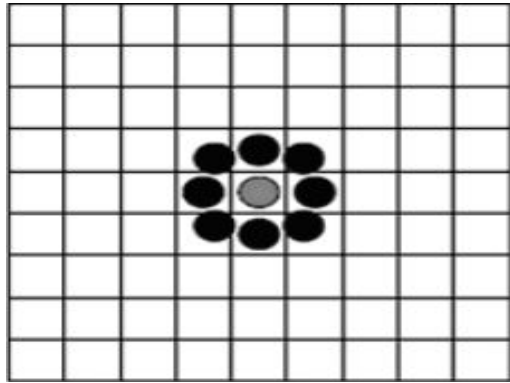
# STANDARD LBP Filter



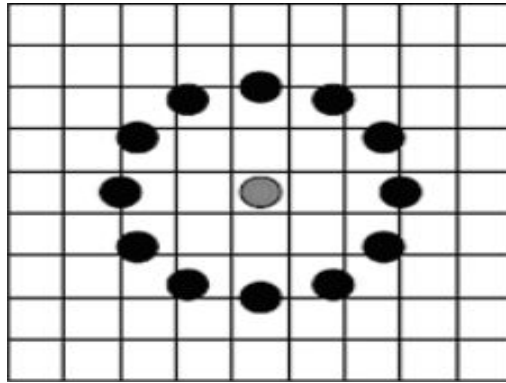
# Advanced LBP<sub>(P,R)</sub>

P = Pixels

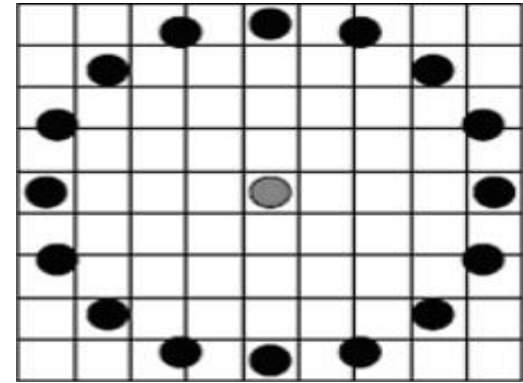
R = Radius



LBP(8,1)



LBP(16,2)



LBP(20,4)

# **L B P   A d v a n t a g e s   a n d   d i s a d v a n t a g e s**

## **Advantages**

- **High discriminative power**
- **Computational simplicity**
- **Invariance to grayscale changes and**
- **Good performance.**

## **Disadvantages**

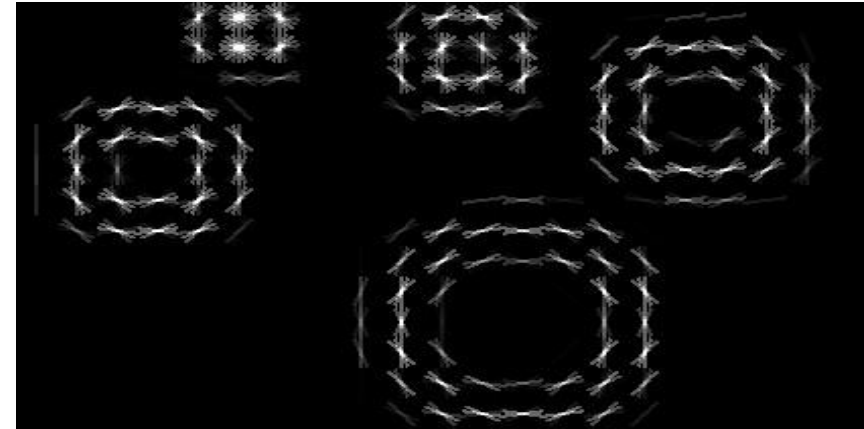
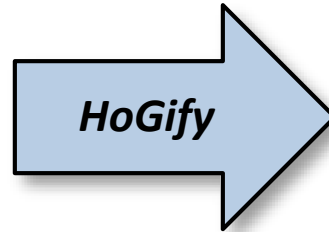
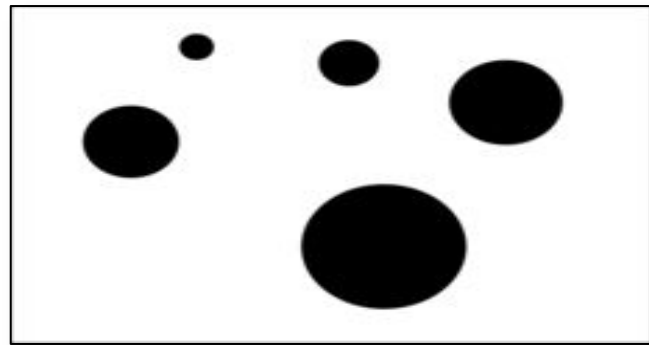
- **Not invariant to rotations**
- **The size of the features increases exponentially with the number of neighbours which leads to an increase of computational complexity in terms of time and space**
- **The structural information captured by it is limited. Only pixel difference is used. magnitude information ignored.**

# **LBP: UNIFORM PATTERNS**

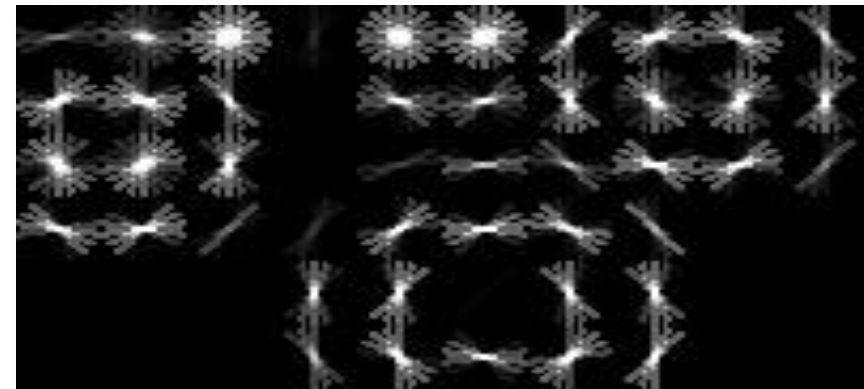
# **Histogram of Oriented Gradient Gradient (HoG)**



# Histogram of Oriented Gradients (HoG)

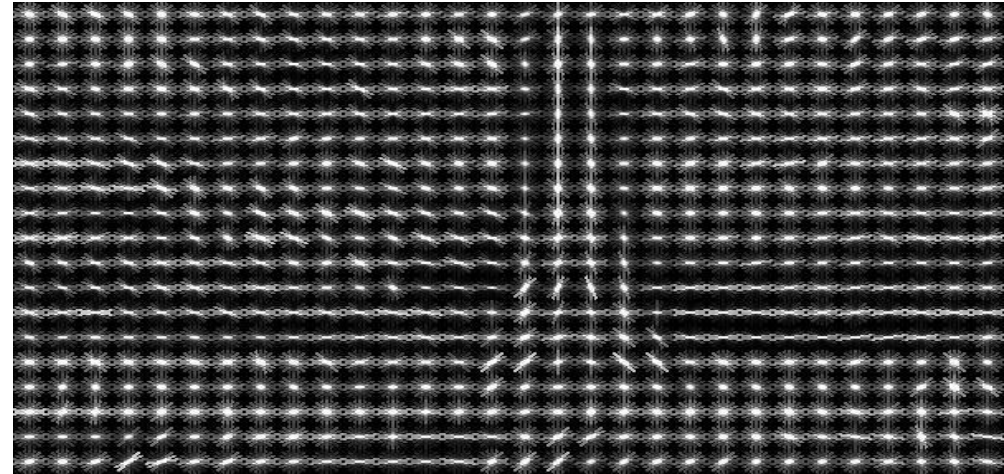


10x10 cells



20x20 cells

# Histogram of Oriented Gradients (HoG)

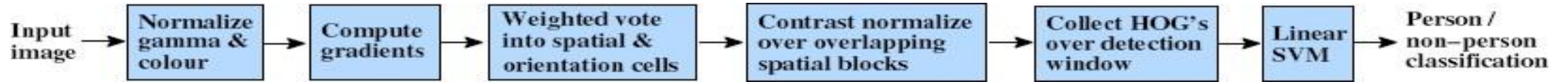


# Histogram of Oriented Gradients (HoG)

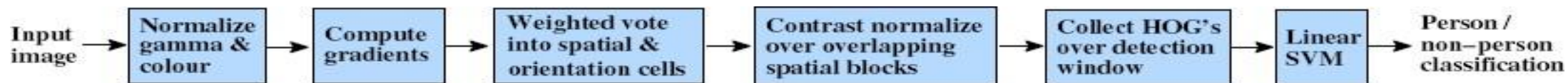
- First used for application of person detection [Dalal and Triggs, CVPR 2005]
- Cited since in thousands of computer vision papers







- Tested with
  - RGB
  - LAB
  - Grayscale
- Gamma Normalization and Compression
  - Square root
  - Log



|    |   |   |
|----|---|---|
| -1 | 0 | 1 |
|----|---|---|

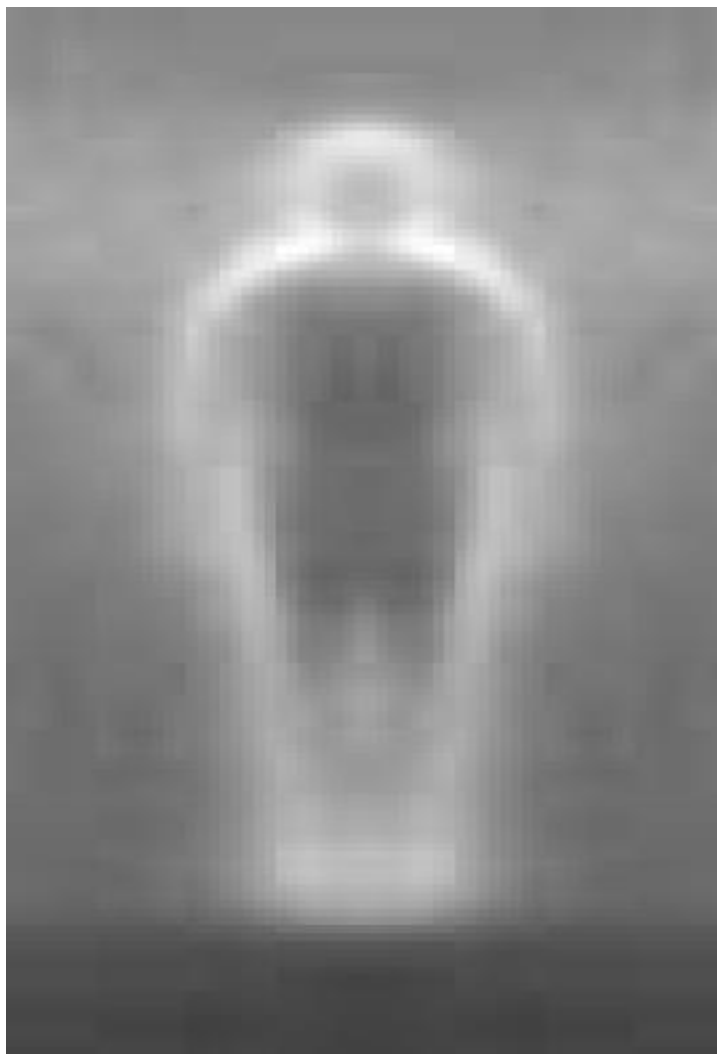
centered

|    |   |
|----|---|
| -1 | 1 |
|----|---|

uncentered

|   |    |   |   |    |
|---|----|---|---|----|
| 1 | -8 | 0 | 8 | -1 |
|---|----|---|---|----|

cubic-corrected



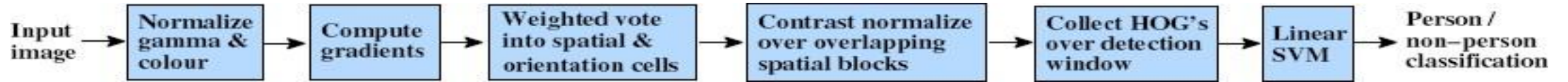
|    |   |
|----|---|
| 0  | 1 |
| -1 | 0 |

diagonal

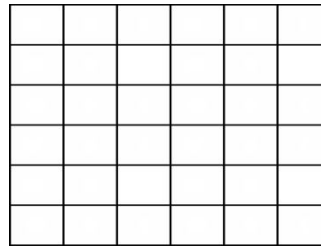
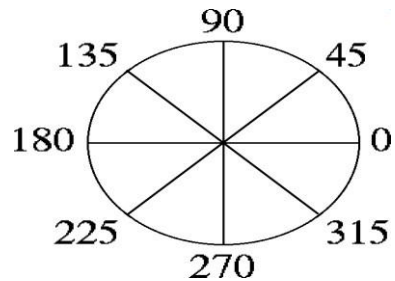
|    |   |   |
|----|---|---|
| -1 | 0 | 1 |
| -2 | 0 | 2 |
| -1 | 0 | 1 |

Sobel

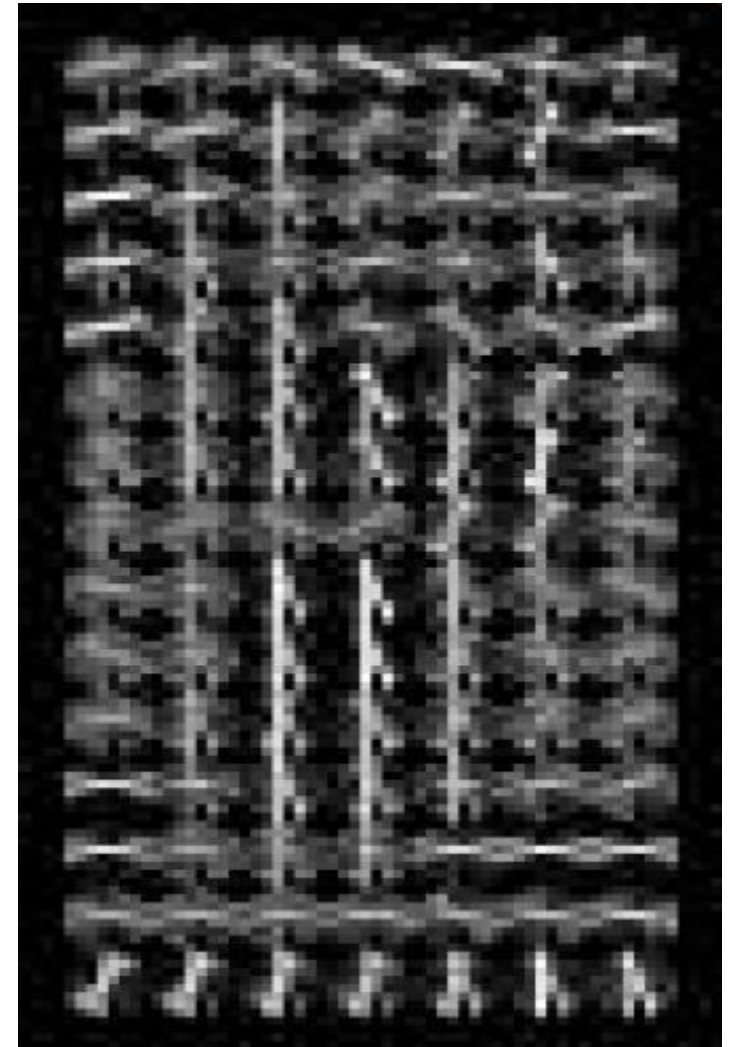




- Histogram of gradient orientations
  - Orientation -Position

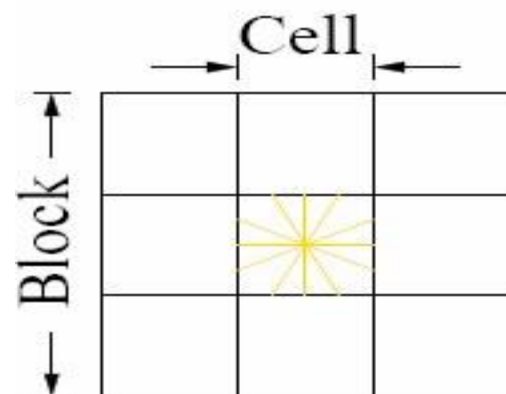


– Weighted by magnitude

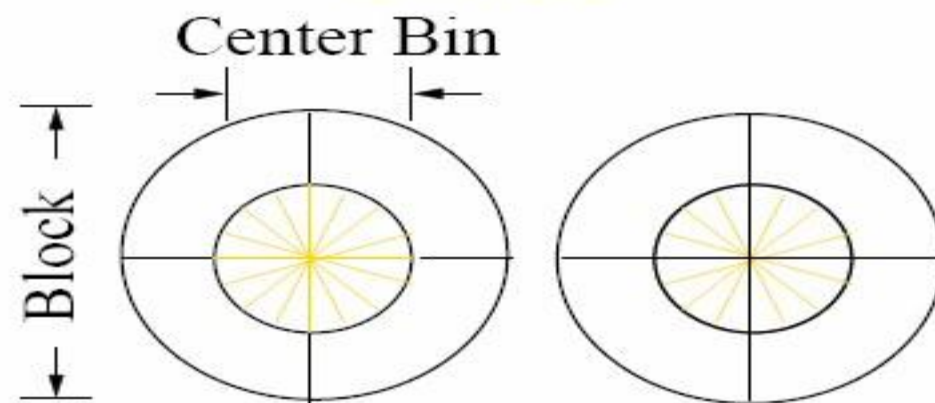




**R-HOG**

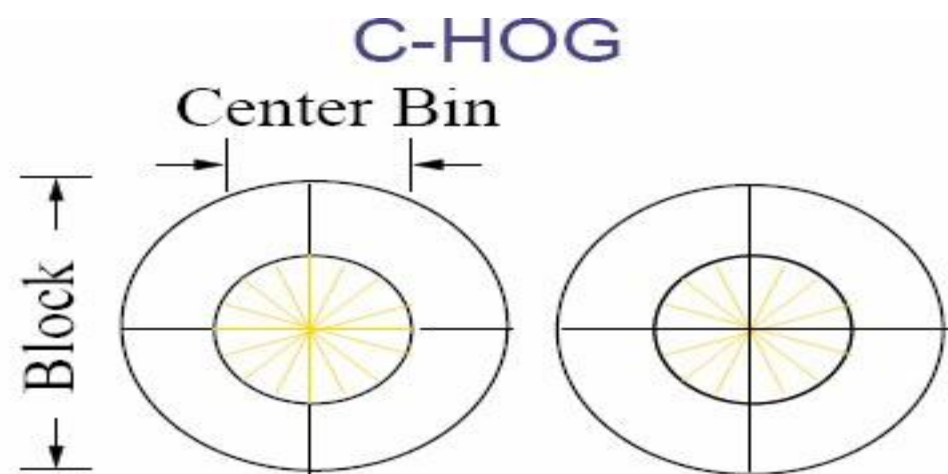
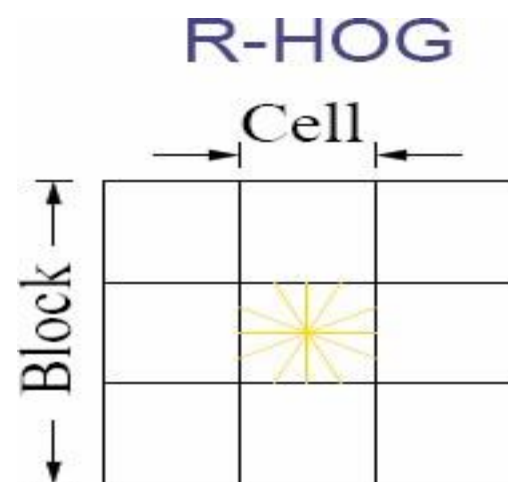


**C-HOG**



Radial Bins, Angular Bins





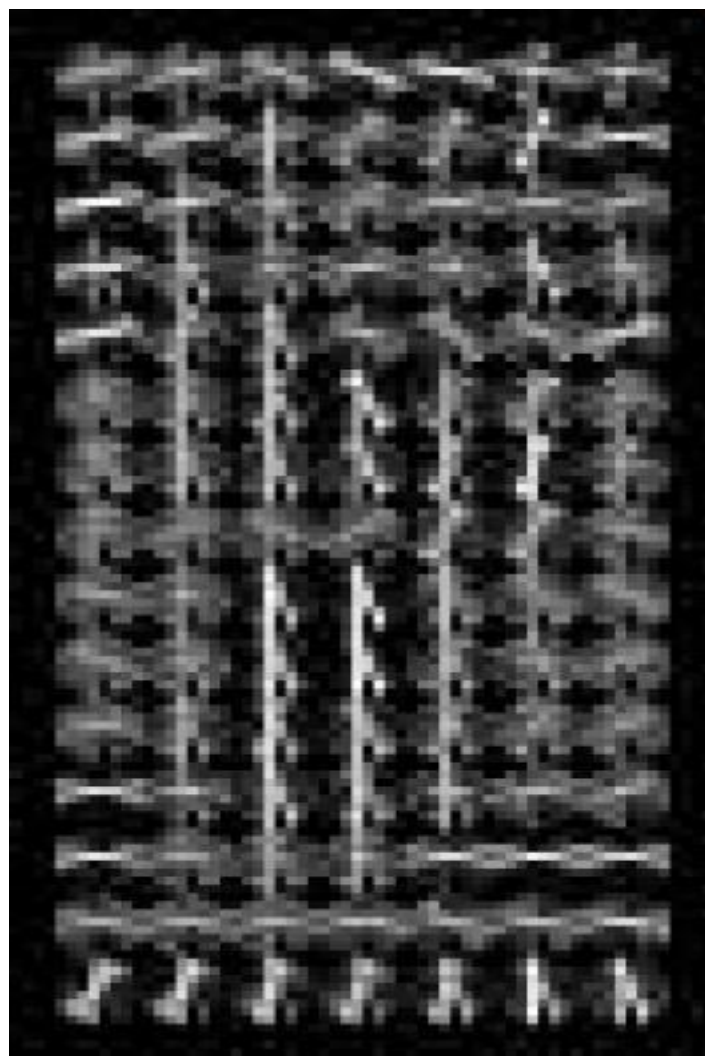
Radial Bins, Angular Bins

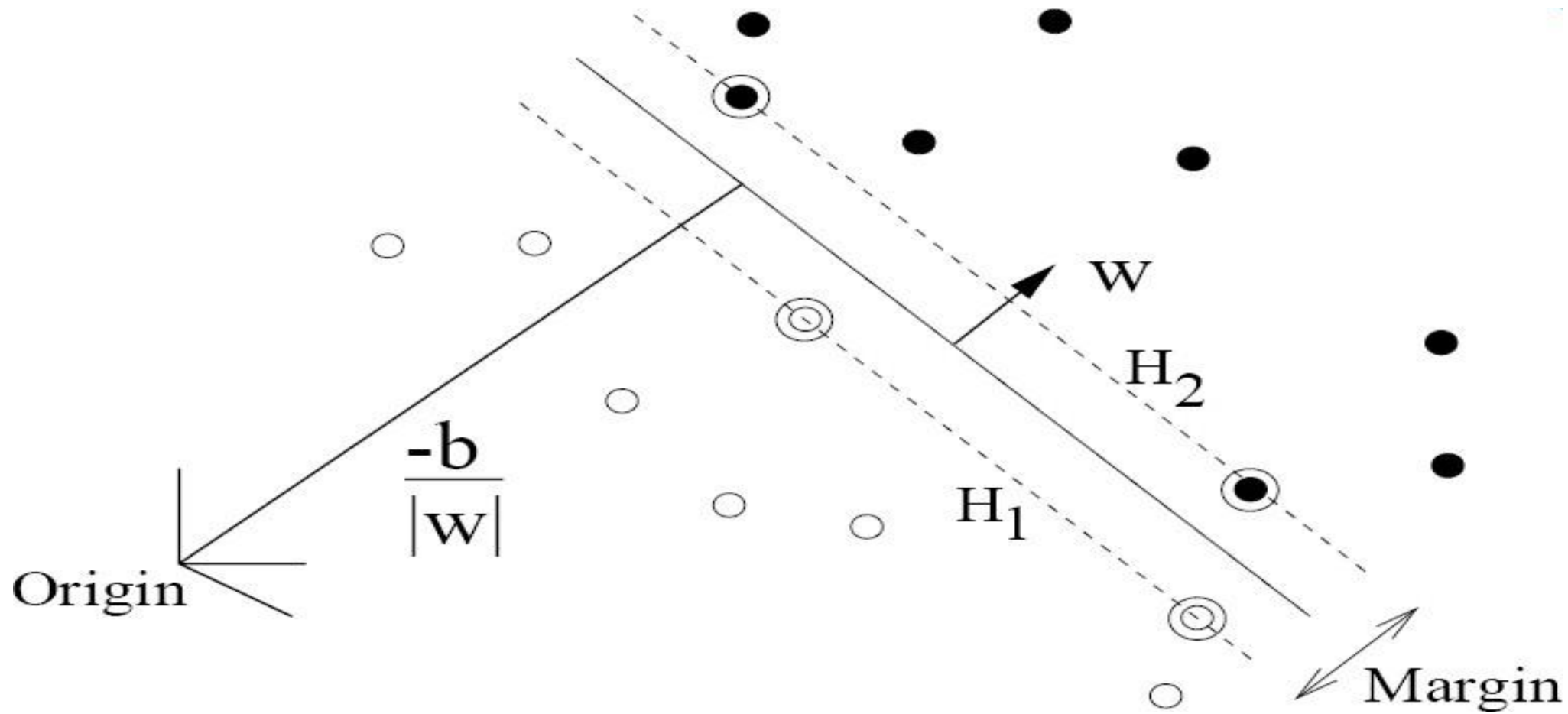
$$L1 - norm : v \longrightarrow v / (\|v\|_1 + \epsilon)$$

$$L1 - sqrt : v \longrightarrow \sqrt{v / (\|v\|_1 + \epsilon)}$$

$$L2 - norm : v \longrightarrow v / \sqrt{\|v\|_2^2 + \epsilon^2}$$

$L2 - hys$  : L2-norm, plus clipping at .2 and renormalizing



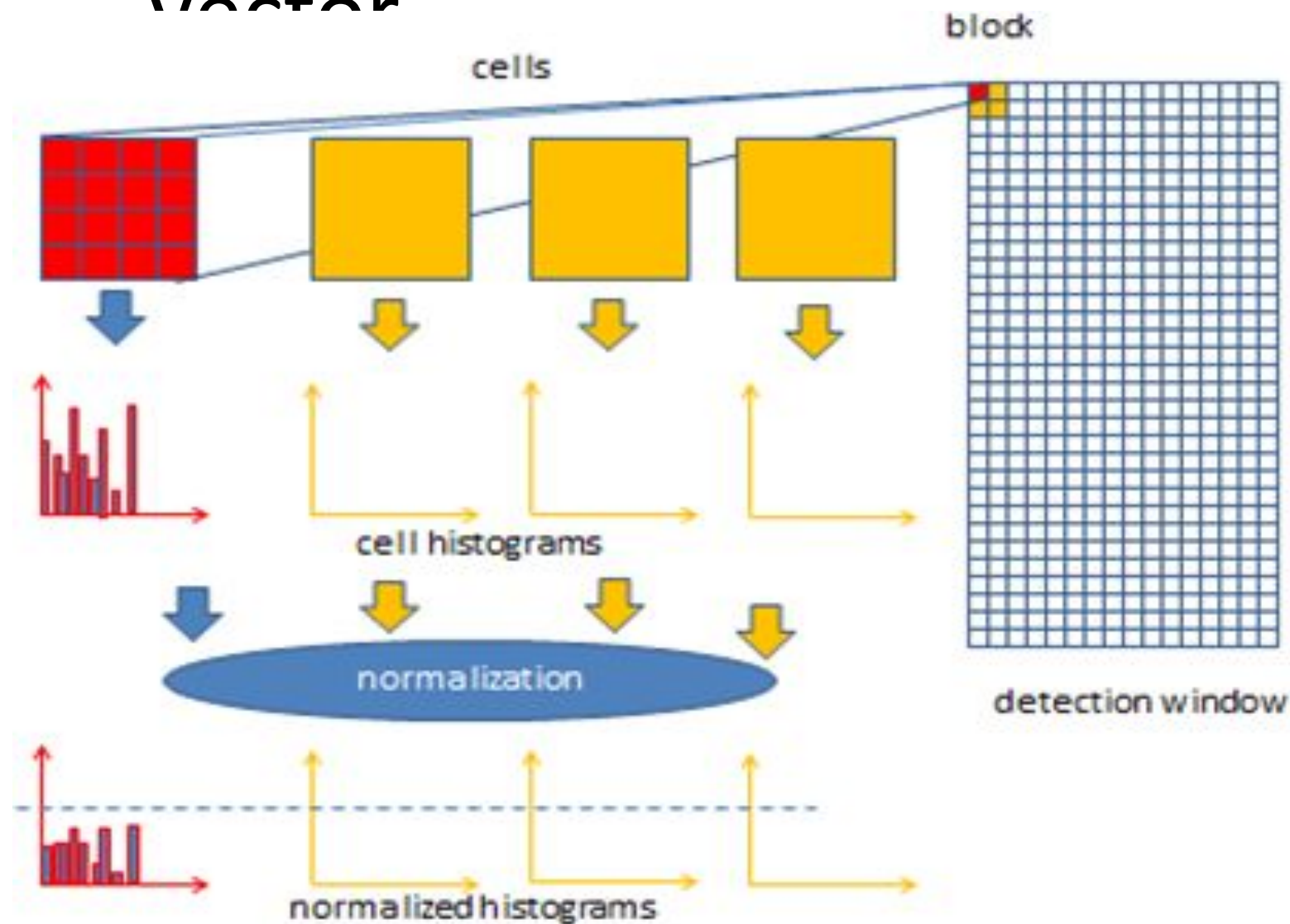








# HoG Based feature Vector



# Acknowledgements

Material in these slides has been taken from, the following resources

- ◆ Digital Image Processing”, Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- ◆ Guo, Zhenhua, Lei Zhang, and David Zhang. "A completed modeling of local binary pattern operator for texture classification." IEEE Transactions on Image Processing 19, no. 6 (2010): 1657-1663.
- ◆ Dalal, Navneet, and Bill Triggs. "Histograms of oriented gradients for human detection." In Computer Vision and Pattern Recognition, 2005. CVPR 2005. IEEE Computer Society Conference on, vol. 1, pp. 886-893. IEEE, 2005.
- ◆ Digital image processing by Dr Usman Akram